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RADIAL NERVE

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I. INTRODUCTION:

The radial nerve is the largest terminal branch of the posterior cord of the brachial plexus. It is a mixed sensory-motor nerve destined for the posterior aspect of the upper limb. It is mainly responsible for upper-limb extension and, secondarily, for supination and thumb abduction.

II. DESCRIPTIVE ANATOMY:

1. **Origin:** It arises from the posterior cord of the brachial plexus in the axillary cavity, behind the pectoralis minor. It contains fibers originating from roots C6–C7–C8 and T1.
2. **Course:** From the posterior and inferior part of the axillary fossa, it descends downward and laterally in the posterior brachial region along the posterior surface of the humerus in the radial groove. It then crosses the lateral intermuscular septum to reach the lateral bicipital groove.
3. **Termination:** It ends at the level of the elbow, opposite the radial head, dividing into two terminal branches: a superficial anterior branch and a deep posterior branch, both continuing to the hand.

III. RELATIONS:

- 1) **In the axillary fossa:** the radial nerve lies on the posterior aspect of the neurovascular bundle, behind the axillary artery.
- 2) **In the arm:** it leaves the lower axillary region through **the humero-tricipital interval** limited by:
 - Superiorly: the tendons of teres major and latissimus dorsi.
 - Laterally: the humerus.
 - Medially: the long head of the triceps brachii.

It then follows an oblique course downward, posteriorly and laterally to reach the posterior surface of the humerus in the osteomuscular canal (radial canal), accompanied by the deep brachial artery located above it. Then it crosses the lateral brachial intermuscular septum.

3) At the elbow: it moves anteriorly, running in the **lateral bicipital groove** bordered by:

- Medially: the biceps tendon and brachialis
- Laterally: brachioradialis and the extensor carpi radialis longus
- Anteriorly: the superficial fascia and brachioradialis
- Posteriorly: the extensor carpi radialis longus and brevis, then the lateral epicondyle.

It lies between brachioradialis laterally and biceps brachii medially. It is accompanied by the radial collateral and radial recurrent arteries, which anastomose.

IV. COLLATERAL BRANCHES:

- Posterior cutaneous nerve of the arm: supplies the skin of the upper third of the posterior arm.
- Nerve to the long head of the triceps
- Superior branch to the medial head of the triceps
- Inferior branch to the medial head: innervates medial head and anconeus.
- Nerve to the lateral head of the triceps
- Inferior lateral cutaneous nerve of the arm: supplies the skin of the lower third of the posterolateral arm.
- Posterior cutaneous nerve of the forearm: supplies the middle portion of the posterior forearm.
- Nerve to brachioradialis.
- Nerve to extensor carpi radialis longus.

V. TERMINAL BRANCHES:

1. Superficial anterior branch:

Purely sensory, It descends vertically within the sheath of brachioradialis, lying laterally to the radial artery in the upper two-thirds of the forearm. In the lower quarter it becomes dorsal and subcutaneous, giving the lateral and medial dorsal digital nerves of the first and second fingers.

2. Deep posterior branch:

Primarily motor. It pierces the supinator muscle between its two heads, emerges posteriorly, and runs between the two muscular layers of the posterior forearm, dividing into its terminal branches.

- Collateral branches: nerve to extensor carpi radialis brevis, nerves to supinator.
- Terminal branches:
 - Muscular branches to the extensor muscles of the fingers.
 - Posterior interosseous nerve.

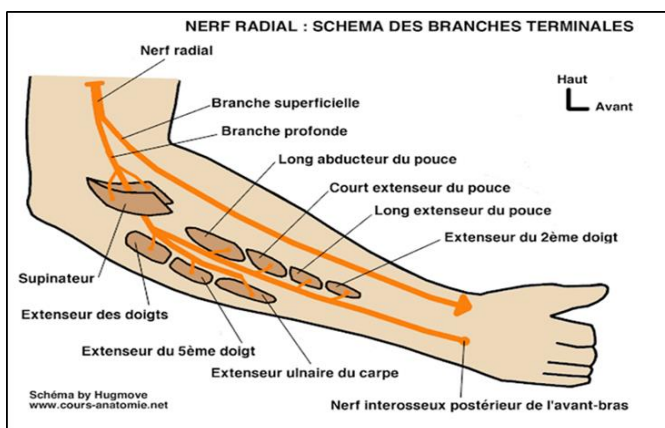
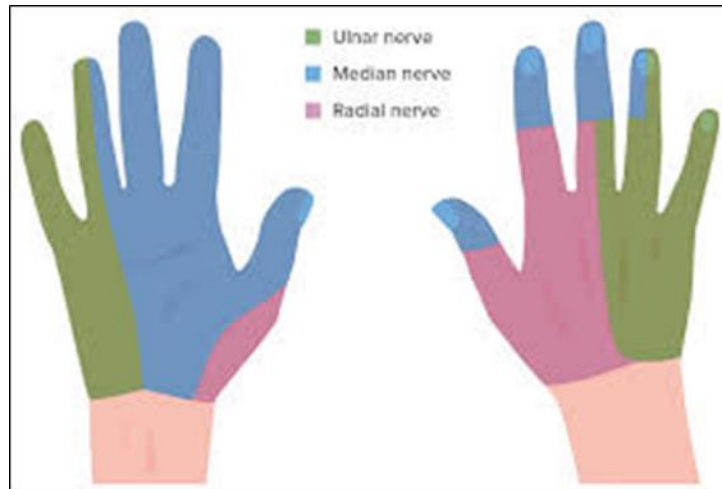
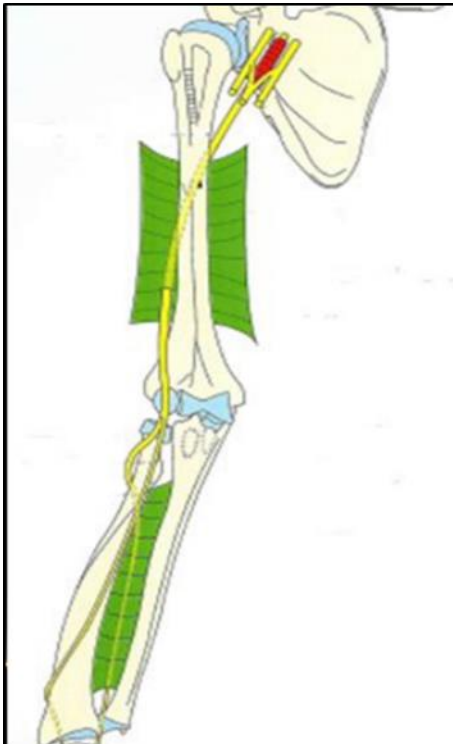
VI. INNERVATION TERRITORY:

- **Motor:** mainly extension of the upper limb; accessory supination and thumb abduction. It innervates the muscles of the posterior arm and the posterior and lateral compartments of the forearm.
- **Sensory:**
 - Posterior arm and posterolateral elbow

- Middle third of the posterior forearm
- Dorsal surface lateral to the axis of the third finger, except the last two phalanges of the second and third fingers
- A small zone at the base of the thenar eminence.

VII.CONCLUSION:

The radial nerve is the most frequently injured nerve of the upper limb. It may be damaged in shoulder dislocation or mid-shaft humeral fractures. Radial nerve palsy results in **“wrist drop”** and atrophy of the posterior forearm.



AXILLARY NERVE

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I. INTRODUCTION:

The axillary nerve is a mixed sensory-motor nerve supplying the shoulder region. It is the nerve responsible for arm abduction. It arises in the axillary cavity from the bifurcation of the posterior secondary trunk of the brachial plexus.

II. DESCRIPTIVE ANATOMY:

- 1. Origin:** In the axillary cavity, behind the pectoralis minor, from the bifurcation of the posterior secondary trunk formed from fibers of C5 and C6.
- 2. Course:** It passes laterally and posteriorly through the lateral axillary space to reach the arm, following a circular course around the surgical neck of the humerus, accompanied by the posterior circumflex humeral artery.
- 3. Termination:** It ends on the deep surface of the deltoid muscle.

III. RELATIONS:

- 1) In the axillary cavity:** it lies on the anterior surface of the subscapularis muscle.
- 2) In the humero-tricipital space** bounded by:
 - Superiorly: subscapularis
 - Inferiorly: teres major
 - Medially: long head of triceps
 - Laterally: surgical neck of the humerus.It is accompanied by the posterior circumflex humeral artery.

IV. COLLATERAL BRANCHES:

- Muscular ramus to subscapularis and teres minor.
- Articular branch to the glenohumeral joint.
- Superior lateral cutaneous nerve of the arm: supplies the skin over the lateral shoulder and arm.

V. TERMINAL BRANCHES:

The nerve terminates as branches entering the deep surface of the deltoid muscle.

VI. INNERVATION TERRITORY:

- **Motor:** abduction and lateral rotation of the arm
- **Sensory:** lateral aspect of the shoulder.

VII. CONCLUSION:

The axillary nerve may be injured in shoulder dislocation or fractures of the surgical neck of the humerus. Injury results in the inability to raise the arm forward, backward, or horizontally, accompanied by deltoid muscle atrophy.

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